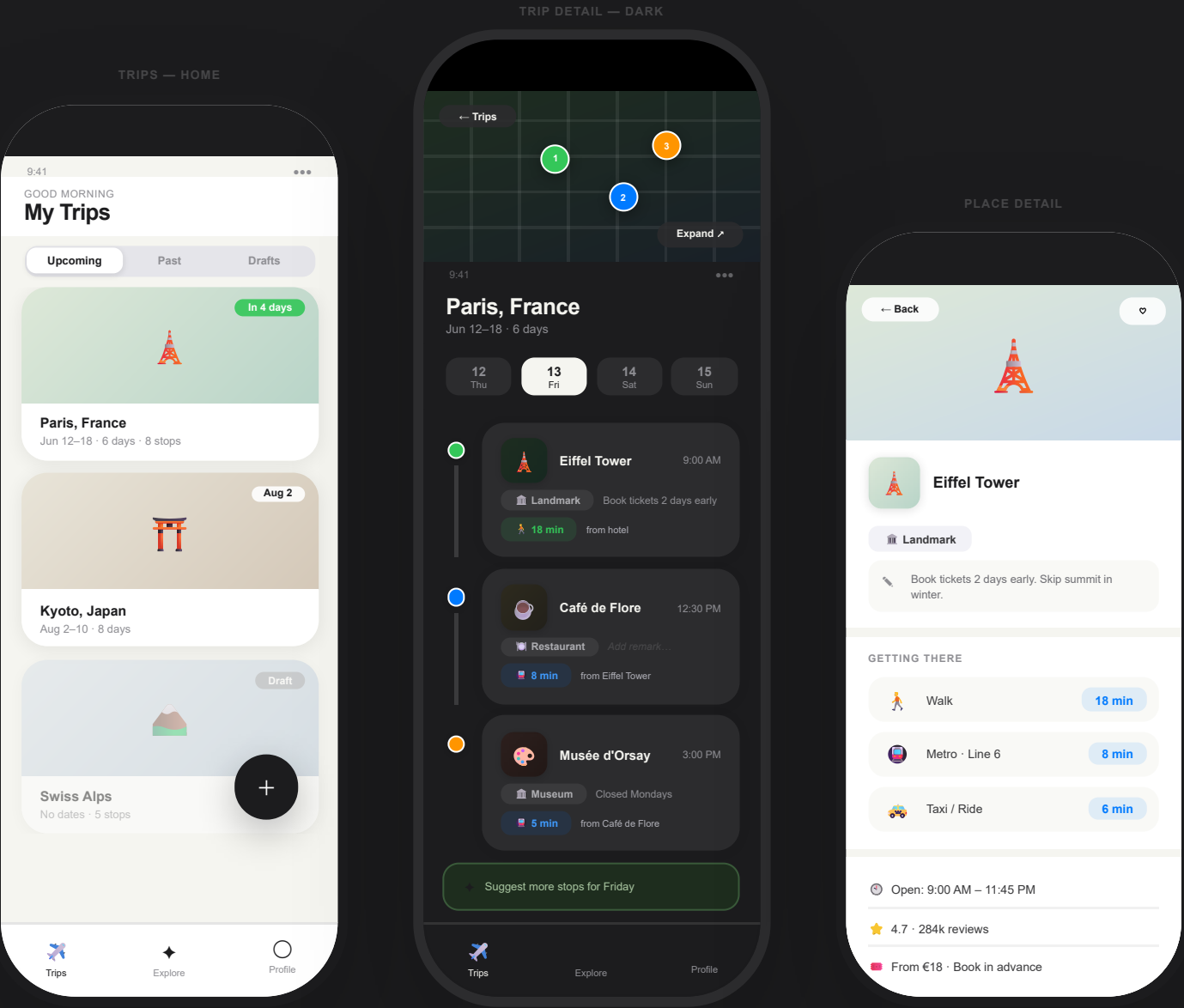


Smart Travel Planner

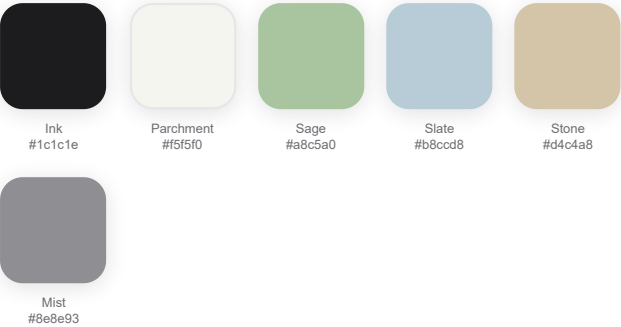
Mobile application system design — architecture, screen design, data model, services, and interaction patterns for a cross-platform iOS & Android travel planning experience.



Design Language

Neo-minimalism + Apple-inspired. Earthy calm palette, system typography, spring-curve animations, frosted glass surfaces.

COLOUR PALETTE



TYPOGRAPHY SCALE

DISPLAY

TITLE

BODY

LABEL

CAPTION

Paris

My Trips

Jun 12–18 · 6 days

UPCOMING TRIP

18 min

GESTURE SYSTEM

Sheet drag
Snap points at 30% / 60% / 95% of screen height

Card swipe left
Reveals Delete / Archive; spring-back on cancel

Long-press drag
Lift stop card; haptic on pickup; spring settle on drop

Pinch map
Native Google Maps zoom; pins cluster at low zoom

Day strip scroll
Horizontal snapping; selected pill auto-centres

Information Architecture

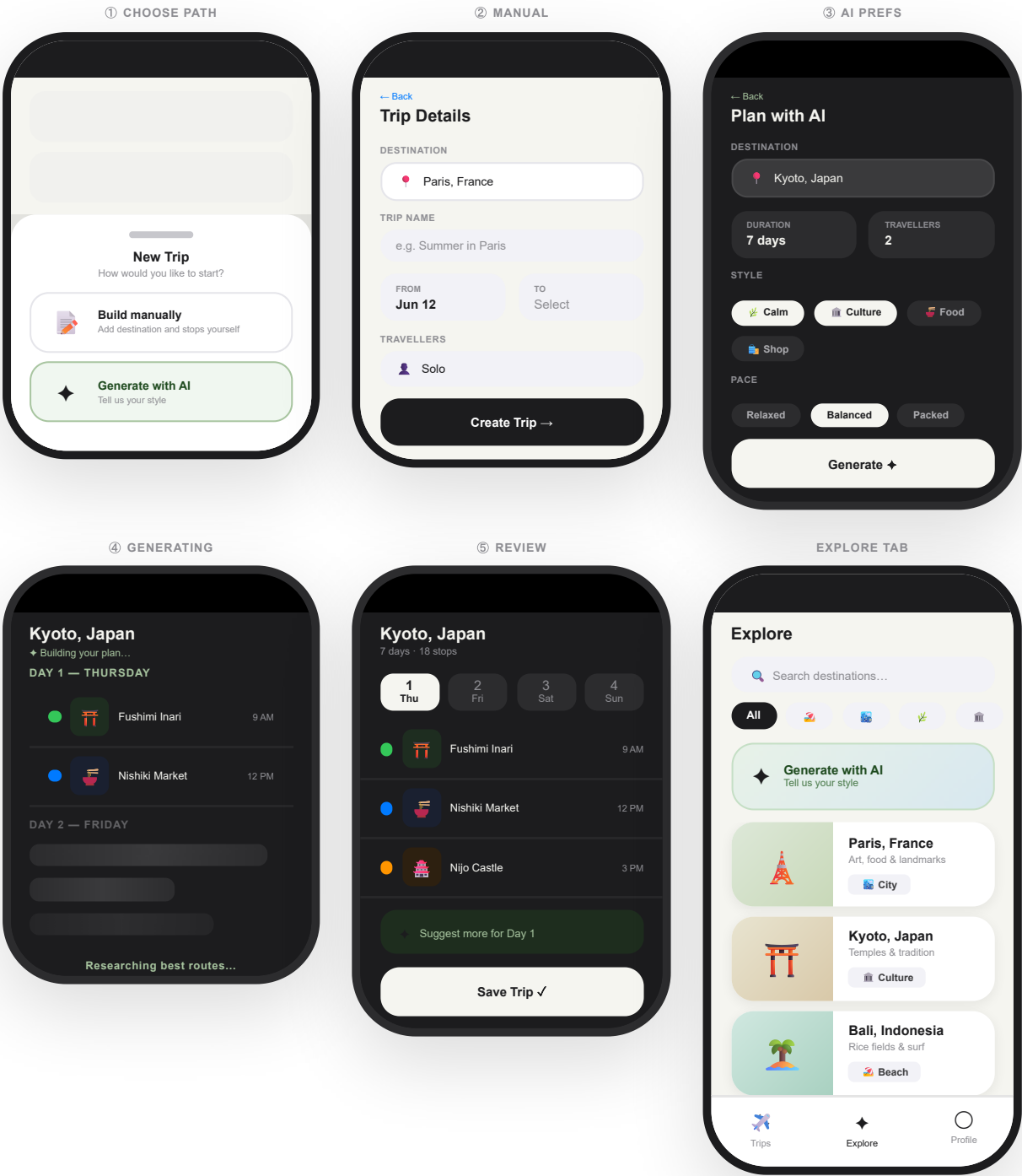
3-tab shell. Map lives inside Trip Detail — not a standalone tab. Onboarding gate before main app.

App

- Onboarding / Auth — Welcome slides · Sign In / Sign Up · Supabase Auth
- Trips** (home tab)
 - Trip List — Upcoming / Past / Drafts · FAB (+)
 - Create Trip — **Manual** (form) | **AI Generate** (style + pace → stream → refine)
 - Trip Detail — Map hero · Day strip · **Activity list (stop cards)** · Add stop · AI suggest
 - Place Detail — Hero · Name+icon · Category · Remark · Transport · Info
 - Explore** — Search · Category filter · AI Generate banner · Destination cards
 - Profile** — Settings · Saved Places · Offline Trips

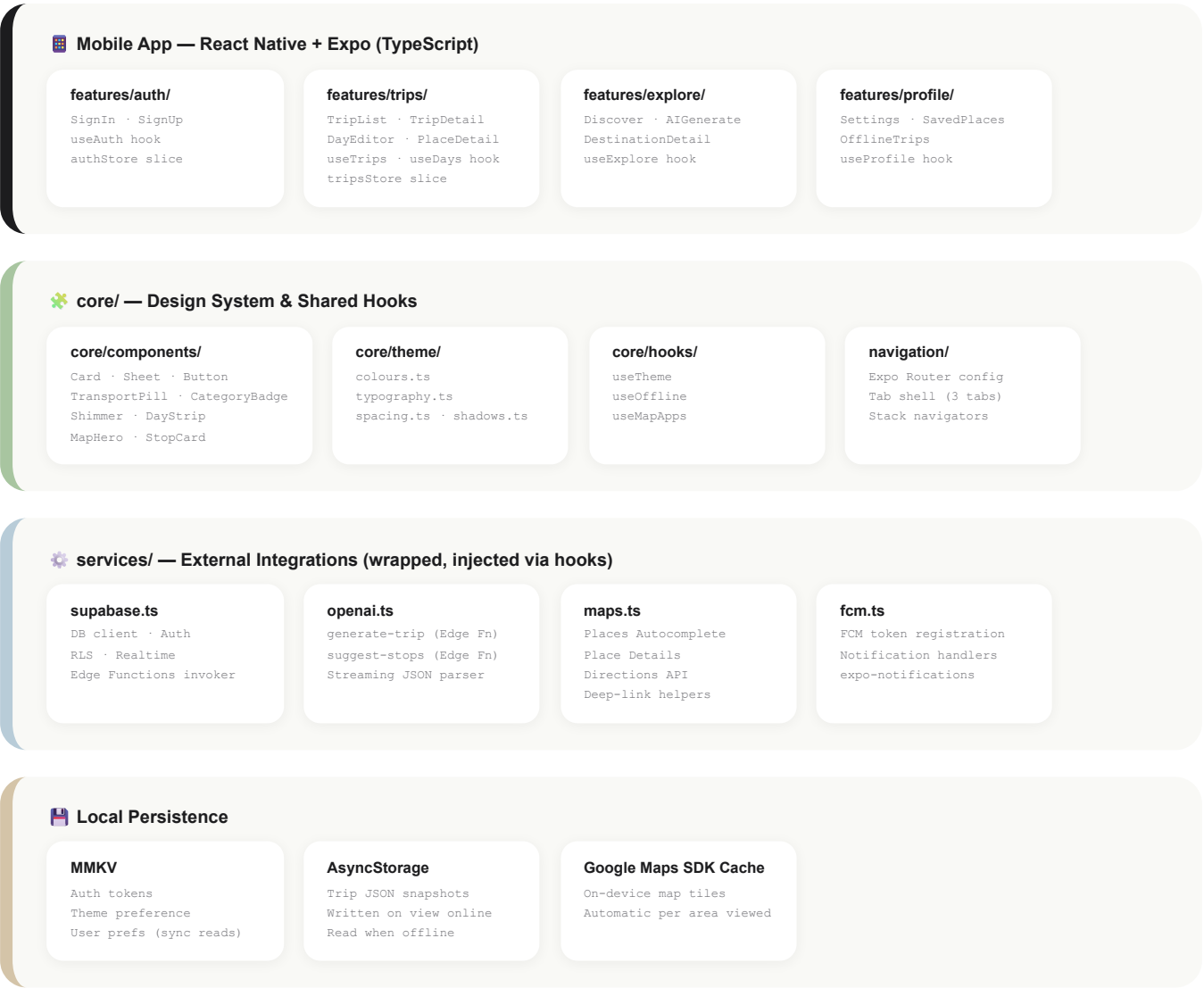
Create Trip Flow & Explore Screen

FAB opens a dual-path sheet. AI path streams itinerary staggered by day. Explore tab has category filters and an AI Generate banner.

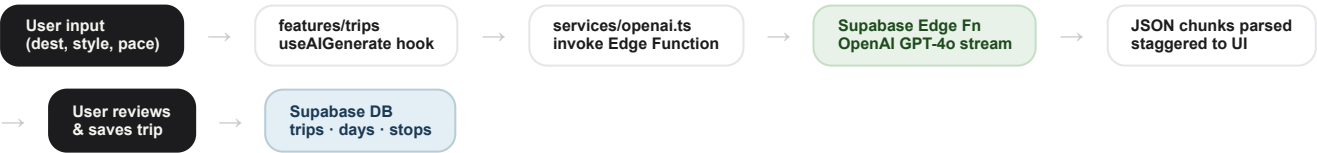


System Architecture

Hybrid Feature + Shared Core pattern. Features are self-contained slices. Services are injected via hooks. Core holds the design system.



DATA FLOW — TRIP CREATION (AI PATH)



Database Schema

Six tables in Supabase / PostgreSQL. Row-level security on all tables. `places` is a shared read-only cache of Google Places data. OpenAI is called server-side only — API key never on device.

 users

PK id	uuid
email	text
display_name	text
avatar_url	text
travel_style	text[]
default_pace	text
created_at	timestampz

 trips

PK id	uuid
FK user_id	uuid
name	text
destination	text
dest_lat / lng	float8
start_date	date
end_date	date
status	enum
cover_emoji	text
ai_generated	bool

 days

PK id	uuid
FK trip_id	uuid
date	date
day_number	int2

 stops

PK id	uuid
FK day_id	uuid
FK place_id	uuid
scheduled_time	time
order_index	int2
remark	text

 places

PK id	uuid
google_place_id	text unique
name	text
lat / lng	float8
category	text
thumbnail_url	text
rating	float4
opening_hours	jsonb
cached_at	timestampz

 transport_segments

PK id	uuid
FK from_stop_id	uuid
FK to_stop_id	uuid
mode	enum
duration_min	int2
route_label	text

RELATIONSHIPS

users	→ 1:many →	trips	user owns trips
trips	→ 1:many →	days	trip spans days
days	→ 1:many →	stops	day has ordered stops
stops	→ many:1 →	places	stops reference cached Google Place data
stops	→ 1:many →	transport_segments	each stop has segments to the next stop

ROW-LEVEL SECURITY

All tables except `places` enforce: users can only SELECT / INSERT / UPDATE / DELETE their own rows (matched by `auth.uid()`). The `places` table is readable by all authenticated users and writable only by the service role (populated by the server when stops are added).

Services, Error Handling & Testing



Supabase

Auth + Database + Realtime

Email + Google OAuth. All CRUD on trips/days/stops/places. RLS enforced. Realtime sync across devices for same user. Edge Functions host the OpenAI calls and push notification scheduler.

`.from('trips').select()``.upsert()``.subscribe()``functions.invoke()`



OpenAI

AI Itinerary Generation + Stop Suggestions

Called server-side via Supabase Edge Function only — API key never on device. GPT-4o with response_format: json_schema streams structured day→stops arrays. "Suggest more" passes existing stops as context.

`generate-trip (Edge Fn)``suggest-stops (Edge Fn)``response_format: json_schema`



Google Maps

Maps · Places · Directions

react-native-maps with Google Maps provider. Places Autocomplete for stop search. Place Details populates the places cache (name, category, thumbnail, hours, rating). Directions API fetches transport durations stored in transport_segments. Places re-fetched if cached_at > 7 days.

`Places Autocomplete``Place Details API``Directions API``Maps SDK`



Firebase FCM

Push Notification Reminders

Trip-day reminders sent via a Supabase scheduled Edge Function each morning. expo-notifications handles FCM token registration and foreground/background display. Users opt out per trip in Profile → Settings.

`Expo.Notifications.getToken()``FCM via Edge Function`



MMKV + AsyncStorage

Local Persistence & Offline Cache

MMKV: auth tokens, theme, prefs (synchronous reads). AsyncStorage: JSON trip snapshots written on every online view, read when offline detected via @react-native-community/netinfo.

`MMKV.set('token', ...)``AsyncStorage.setItem('trip:id', JSON)`

ERROR HANDLING

SCENARIO	HANDLING
Offline No network	Sage banner slides in. AsyncStorage snapshot loads trips. Network-only features grey out with tooltip.
API OpenAI fails	Inline "Couldn't generate — try again" with retry button. No modal, no crash.
API Maps unavailable	Transport pills show "-". Map hero shows static placeholder. Search shows "unavailable".
Auth Token expired	Supabase auto-refreshes. On failure, quiet redirect to Sign In with "Session expired" notice.
User Remark save fails	Optimistic update shown; on failure, value reverts + red underline + retry tap.
Offline Map app missing	Linking.canOpenURL() checked first; only installed apps shown; browser fallback if none.

TESTING STRATEGY

UNIT — JEST

- Zustand store actions
- Service helpers
- AI JSON parser
- Offline cache logic

COMPONENT — RTL

- Stop card renders correctly
- Transport pill tap
- Day strip scroll
- Remark save on blur

INTEGRATION — SUPABASE LOCAL

- Create trip/day/stop flow
- RLS enforcement
- Stop reorder persistence
- Transport segment creation

E2E — MAESTRO

- Sign up → create → stop → detail
- AI generate flow
- Offline trip load
- Dark mode persistence